Test Case ID	Test Focus / Operation	Input / Execution Path	Legacy Outcome (Before Refactoring)	Refactored Outcome (After Refactoring)	Regression Status & Verification Observation
TC-01	Basic Element Insertion	Insert 6 elements (10, 20, 30, 40, 50, 60)	Functional in global scope; risk of unintended mutation.	Functional and encapsulated inside DynamicArray object.	PASSED — Data structures isolated from global scope.
TC-02	Dynamic Capacity Resizing	Insert elements exceeding initial capacity (capacity = 5)	Memory Leak; old memory buffer orphaned without delete[].	Zero Memory Leaks; old buffer freed via delete[] temp.	PASSED — Heap memory usage remains stable.
TC-03	Full Traversal Output	Loop iteration to display list contents	Off-by-one heap over-read (i <= currentCount); outputs garbage value.	Clean traversal (0 ... currentCount - 1); accurate boundary access.	PASSED — Garbage values and buffer over-reads eliminated.
TC-04	Linear Search Target	Search for target value 30	Located target at index 2 correctly.	Located target at index 2 correctly.	PASSED — Behavioral regression verified; functionality intact.
TC-05	Linear Search Non-Existent	Search for target value 99	Returned -1 index safely.	Returned -1 index safely.	PASSED — Correct handling of missing element searches.
TC-06	Element Deletion Shifting	Delete element at index 2 (value 30)	Elements shifted left using redundant nested for loop.	Elements shifted left using single-pass O(n) loop.	PASSED — Streamlined shifting logic without algorithmic overhead.
TC-07	Deletion at Lower Bound	Delete element at index 0	Shifted remaining items left; initial capacity untouched.	Shifted remaining items left; size updated cleanly.	PASSED — Lower boundary deletion verified.
TC-08	Upper Boundary Check	Attempt removal at invalid index 99	Invalid memory write / system crash due to missing bounds check.	Bounds check catches error; displays "Error: Invalid index 99!" cleanly.	PASSED — System protected against out-of-bounds indices.
TC-09	Matrix Transpose Processing	Process 3x3 grid transposition	Transposed matrix printed, but limited by static dimension assumptions.	Transposed matrix printed raw without modifying raw data buffers.	PASSED — Matrix operation verified functional.
TC-10	System Program Cleanup	Exit main application lifecycle	Leaked heap array memory completely upon process exit.	Destructor ~DynamicArray( ) executes delete[] dataArray cleanly.	PASSED — 100% heap allocation released back to system.